

Topic

- What is Hierarchical Clustering
 - Agglomerative approach
 - Dendrogram concept
 - Difference from K-Means
 - Practical implementation
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Class Objective

Students will:

- Understand **hierarchical clustering concept**
 - Learn how clusters are formed step-by-step
 - Understand **dendrogram (visual tree)**
 - Apply clustering using **Agglomerative Clustering**
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1. CLASS START (Hook)

Start like this:

“In K-Means, we decided K first...

But what if we don't want to decide K in the beginning?”

 Pause

Then:

“Today we learn a method where clusters are built step-by-step — Hierarchical Clustering.”

2. WHAT IS HIERARCHICAL CLUSTERING?

Definition

Hierarchical Clustering builds clusters in a **tree-like structure**.

Simple Meaning

“It groups data step-by-step like a family tree.”

3. TYPES (Mention Only)

◆ Agglomerative (Bottom-Up) (Focus)

- Start with individual points

- Merge closest clusters

◆ Divisive (Top-Down)

- Start with one big cluster
- Split into smaller clusters

👉 Focus only on **Agglomerative**

4. ⚙️ HOW AGGLOMERATIVE WORKS

Step-by-Step

1. Start: each point = separate cluster
 2. Find closest clusters
 3. Merge them
 4. Repeat until all points merged
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💡 Teaching Line

“Clusters are built by merging similar points step-by-step.”

5. 🌳 WHAT IS DENDROGRAM?

Definition

A dendrogram is a tree diagram that shows how clusters are merged.

Simple Meaning

“It shows the history of cluster formation.”

6. 📊 HOW TO READ DENDROGRAM

- Bottom → individual points
 - Top → one big cluster
 - Height → distance between clusters
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👉 Cut the tree at a level → get clusters

💡 Teaching Line

“Cut the dendrogram to decide number of clusters.”

7. 🇮🇹 DIFFERENCE: K-MEANS vs HIERARCHICAL

Feature	K-Means	Hierarchical
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Need K before	Yes	No
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Approach	Partition	Tree-based
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Flexibility	Less	More
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Speed	Faster	Slower
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8. ⚠️ LIMITATIONS

✗ Slow for large data

✗ Hard to scale

✗ Once merged, cannot undo

9. 🎯 INTERVIEW CONCEPTS

Q1: What is hierarchical clustering?

👉 Tree-based clustering.

Q2: What is dendrogram?

👉 Tree diagram showing cluster merging.

Q3: Difference from K-Means?

👉 K-Means needs K first, hierarchical doesn't.

10. 💻 PRACTICAL (YOUR FORMAT)

🎯 Objective

12. OUTPUT INTERPRETATION

- Points grouped step-by-step
 - Dendrogram shows merging process
 - Cutting tree gives clusters
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Example:

- Two clear clusters visible
- “Hierarchical clustering builds clusters step-by-step and gives us a full picture of how data is grouped — making it very useful for understanding data structure.”